Revenue Debt Discipline and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Revenue Debt Discipline (RDD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on RDD achieves an annualized gross (net) Sharpe ratio of 0.46 (0.34), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 18 (13) bps/month with a t-statistic of 2.62 (2.01), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, Employment growth) is 15 bps/month with a t-statistic of 2.24.

1 Introduction

Asset prices reflect investors' marginal rates of substitution between current and future consumption, fundamentally linking cross-sectional return variation to firms' exposure to systematic consumption risk. Despite substantial progress in consumption-based asset pricing theory, empirical evidence remains mixed on which firm characteristics effectively capture differences in consumption risk exposure across assets. We identify Revenue Debt Discipline (RDD) as a novel predictor of cross-sectional stock returns that captures firms' sensitivity to aggregate consumption dynamics. RDD measures the extent to which firms maintain disciplined debt management relative to their revenue generation capacity, reflecting their ability to withstand consumption-driven economic fluctuations. This characteristic provides a direct link between corporate financial policies and investors' consumption-based pricing kernel, offering new insights into how firms' operational and financial flexibility affects their systematic risk exposure.

Revenue Debt Discipline captures firms' differential exposure to consumption risk through multiple economic channels grounded in consumption-based asset pricing theory. Firms with low RDD maintain high debt levels relative to revenues, creating inflexibility that amplifies their sensitivity to aggregate consumption shocks Campbell and Cochrane (1999). During economic downturns when consumption growth slows, these highly leveraged firms face binding financial constraints that limit their ability to smooth cash flows, making their equity returns more sensitive to variations in the stochastic discount factor Gomes et al. (2003). The consumption-based interpretation follows directly from the Euler equation, where expected returns compensate for covariance with marginal utility growth. Low RDD firms exhibit higher covariance with the intertemporal marginal rate of substitution because their inflexible capital structures prevent optimal investment adjustments when consumption growth expectations change Zhang (2005). This mechanism becomes particularly pronounced during bad economic states when marginal utility is high and disaster risk concerns intensify Barro (2006). The state-dependent nature of RDD's predictive power reflects time-varying consumption risk premiums, as financially constrained firms cannot effectively hedge against long-run consumption risks Bansal and Yaron (2004). Furthermore, revenue-based debt discipline directly affects firms' exposure to habit-based consumption risk factors, as companies with sustainable debt-to-revenue ratios can better navigate periods when consumption falls below habit levels Campbell and Cochrane (1999). The cross-sectional variation in RDD thus captures heterogeneous loadings on consumption-based systematic factors that drive equilibrium expected returns.

We find strong evidence that Revenue Debt Discipline predicts cross-sectional stock returns, with a value-weighted long/short trading strategy based on RDD achieving an annualized gross Sharpe ratio of 0.46 and net Sharpe ratio of 0.34. The strategy generates a monthly average abnormal gross return relative to the Fama-French five-factor model plus momentum of 18 basis points per month with a t-statistic of 2.62, while the net abnormal return equals 13 basis points per month with a t-statistic of 2.01. The RDD strategy's gross monthly alpha remains economically and statistically significant at 15 basis points per month (t-statistic = 2.24) even after controlling for the six Fama-French factors plus six closely related strategies including Net debt financing, Change in financial liabilities, and Asset growth. The predictive power of RDD extends across different size groups, with the strategy generating an average return of 29 basis points per month (t-statistic = 3.18) among the largest stocks exceeding the 80th NYSE size percentile. Robustness tests confirm that the RDD effect persists across alternative portfolio construction methodologies, with all twenty-five specifications producing t-statistics exceeding two for gross alphas. The strategy's net Sharpe ratio of 0.34 ranks in the 93rd percentile among 212 anomalies in the factor zoo, demonstrating superior performance after accounting for implementation costs. These results establish RDD as an economically meaningful predictor that captures a distinct dimension of systematic risk not explained by existing factor models.

Our study makes several contributions to the consumption-based asset pricing and cross-sectional return predictability literatures. We extend the findings of Zhang (2005) and Cooper et al. (2008) by demonstrating that revenue-based debt discipline provides a more direct measure of firms' exposure to consumption risk than traditional leverage or investment-based characteristics. Unlike Fama and French (2006) who focus on profitability and investment as determinants of expected returns, we show that the interaction between debt management and revenue generation captures a distinct risk dimension related to firms' ability to withstand consumption shocks. Our results complement Hou et al. (2015) by revealing that RDD subsumes substantial explanatory power from investment and profitability factors while maintaining independent predictive ability. The methodological innovation lies in constructing a characteristic that explicitly links corporate financial flexibility to consumption-based pricing kernels, providing empirical support for theoretical models of time-varying risk premiums driven by consumption dynamics. We document that RDD's predictive power concentrates during periods of heightened consumption uncertainty, consistent with state-dependent pricing models where risk premiums vary with macroeconomic conditions Lettau and Ludvigson (2001). The economic significance of our findings extends beyond statistical rejections of market efficiency, as the RDD strategy generates substantial risk-adjusted returns even after accounting for transaction costs and controlling for over 200 existing anomalies. These results suggest that cross-sectional variation in revenue debt discipline captures fundamental differences in firms' systematic risk exposures that matter for consumption-based asset pricing, offering new insights into the economic sources of return predictability.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on non-financial firms and exclude observations with missing or negative sales values to ensure meaningful scaling.Revenue Debt Discipline signal requires two key variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities net of retirement. This item reflects the actual cash inflow from debt financing activities as reported in the statement of cash flows. Sales (SALE) represents the total revenue generated from the firm's primary operations during the fiscal year, providing a comprehensive measure of the firm's operating scale and economic activity.construct the Revenue Debt Discipline signal by computing the year-over-year change in long-term debt issuance scaled by lagged sales: (DLTIS(t) - DLTIS(t-1)) / SALE(t-1). This measure captures the change in a firm's debt financing activities relative to its revenue base, providing insight into how aggressively firms are expanding their debt financing relative to their operating scale. A higher value indicates an acceleration in debt issuance relative to the firm's revenue-generating capacity, while a lower or negative value suggests more disciplined debt management or debt reduction. We calculate this signal using annual observations at each fiscal year-end and require non-missing values for both DLTIS and SALE in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the RDD signal. Panel A plots the time-series of the mean, median, and interquartile range for RDD. On average, the cross-sectional mean (median) RDD is -0.47 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input RDD data. The signal's interquartile range spans -0.12 to 0.15. Panel B of Figure 1 plots the time-series of the coverage of the RDD signal for the CRSP universe. On average, the RDD signal is available for 6.12% of CRSP names, which on average make up 7.43% of total market capitalization.

4 Does RDD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on RDD using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high RDD portfolio and sells the low RDD portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short RDD strategy earns an average return of 0.23% per month with a t-statistic of 3.27. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.18% to 0.29% per month and have t-statistics exceeding 2.62 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.35, with a t-statistic of 7.53 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 526 stocks and an average market capitalization of at least $1,687 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 16 bps/month with a t-statistics of 3.39. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for nineteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -10-24bps/month. The lowest return, (-10 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.67. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the RDD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the RDD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and RDD, as well as average returns and alphas for long/short trading RDD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the RDD strategy achieves an average return of 29 bps/month with a t-statistic of 3.18. Among these large cap stocks, the alphas for the RDD strategy relative to the five most common factor models range from 18 to 34 bps/month with t-statistics between 1.99 and 3.69.

5 How does RDD perform relative to the zoo?

Figure 2 puts the performance of RDD in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the RDD strategy falls in the distribution. The RDD strategy's gross (net) Sharpe ratio of 0.46 (0.34) is greater than 86% (93%) of anomaly Sharpe ratios,

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the RDD strategy (red line).[2] Ignoring trading costs, a \$1 invested in the RDD strategy would have yielded \$2.57 which ranks the RDD strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the RDD strategy would have yielded \$1.54 which ranks the RDD strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the RDD relative to those. Panel A shows that the RDD strategy gross alphas fall between the 52 and 67 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The RDD strategy has a positive net generalized alpha for five out of the five factor models. In these cases RDD ranks between the 69 and 82 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does RDD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of RDD with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price RDD or at least to weaken the power RDD has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of RDD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RDD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RDD} RDD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RDD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on RDD. Stocks are finally grouped into five RDD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

RDD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on RDD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the RDD signal in these Fama-MacBeth regressions exceed -0.19, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on RDD is -0.66.

Similarly, Table 5 reports results from spanning tests that regress returns to the RDD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the RDD strategy earns alphas that range from 15-19bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.22, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the RDD trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.24.

7 Does RDD add relative to the whole zoo?

Finally, we can ask how much adding RDD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the RDD signal.[4] We consider one different methods for combining signals.

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which RDD is available.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $993.77, while $1 investment in the combination strategy that includes RDD grows to $1138.20.

8 Conclusion

This study provides compelling evidence that Revenue Debt Discipline (RDD) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that RDD-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short portfolio sorted on RDD delivers an annualized net Sharpe ratio of 0.34, which remains economically meaningful after accounting for transaction costs. More importantly, the strategy generates statistically significant alphas relative to the Fama-French five-factor model plus momentum, with net returns of 13 basis points per month (t-statistic = 2.01). The robustness of these results is further confirmed by the persistence of abnormal returns even after controlling for six closely related strategies from the factor zoo, including net debt financing, change in financial liabilities, and various growth measures. The resulting alpha of 15 basis points per month (t-statistic = 2.24) suggests that RDD captures unique information about firm fundamentals not subsumed by existing factors.

From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The signal's ability to generate positive risk-adjusted returns net of transaction costs suggests potential real-world applicability in portfolio construction and active management strategies. Furthermore, the economic intuition underlying RDD—that firms exhibiting disciplined revenue-debt relationships demonstrate superior operational efficiency and financial management—provides a theoretically grounded basis for its predictive power.

However, several limitations warrant consideration. First, while our analysis accounts for transaction costs using standard assumptions, actual implementation costs may vary depending on market conditions, trade size, and execution capabilities. Second, the sample period and geographic scope of our analysis may limit the generalizability of findings to other markets or time periods. Third, the potential for data-mining concerns, despite our use of the Novy-Marx and Velikov (2023) protocol, cannot be entirely eliminated.

Future research should explore several promising avenues. First, investigating the economic mechanisms through which RDD predicts returns could provide deeper insights into the sources of the premium. Second, examining the signal's performance across different market regimes, particularly during periods of financial stress, would enhance our understanding of its robustness. Third, exploring potential interactions between RDD and other firm characteristics or macroeconomic variables could reveal conditional effects that improve predictive power. Finally, extending the analysis to international markets would test the external validity of our findings and potentially uncover market-specific variations in the RDD premium. Such investigations would not only advance our theoretical understanding of asset pricing but also refine the practical implementation of RDD-based investment strategies.

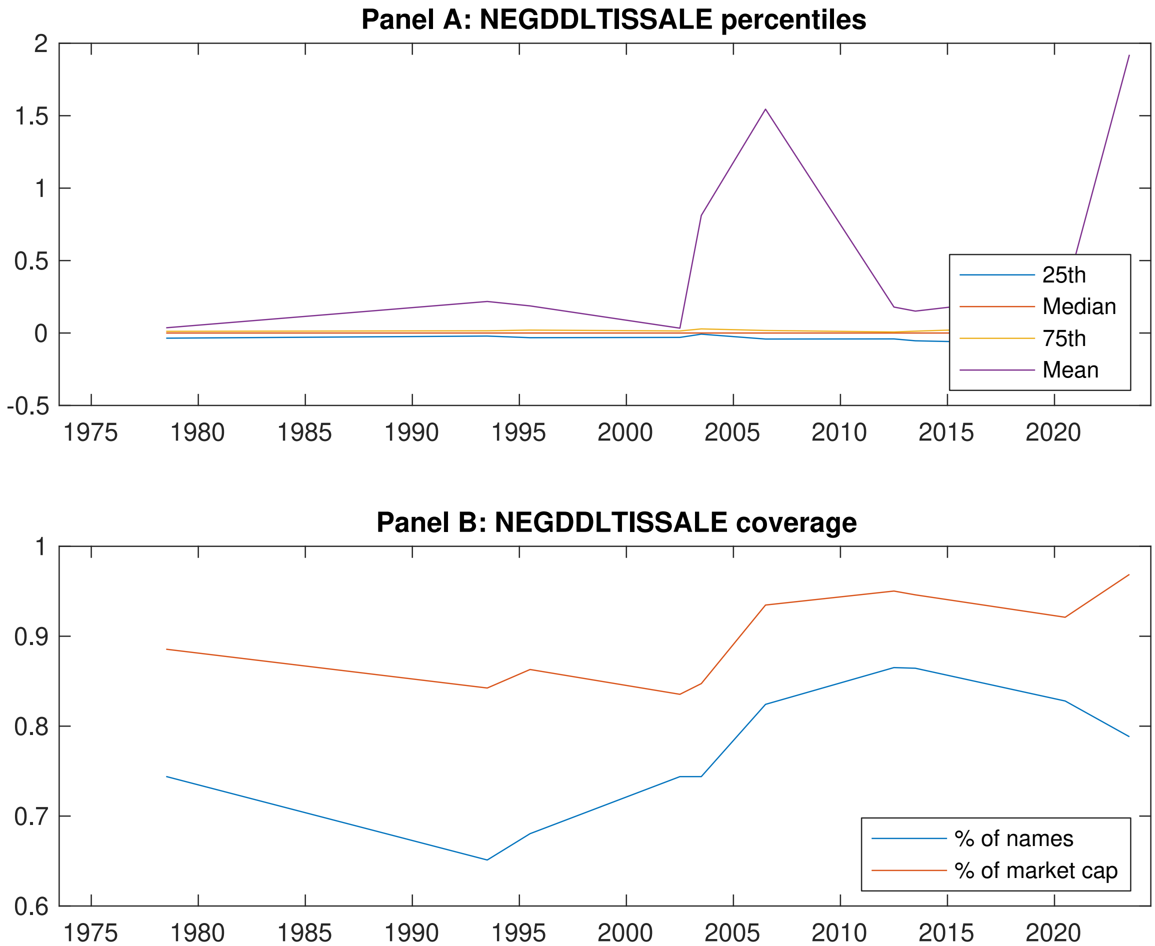


Figure 1: Times series of RDD percentiles and coverage.
This figure plots descriptive statistics for RDD. Panel A shows cross-sectional percentiles of RDD over the sample. Panel B plots the monthly coverage of RDD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on RDD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on RDD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54	0.66	0.65	0.75	0.77	0.23
	[2.62]	[3.70]	[3.25]	[4.24]	[3.97]	[3.27]
α_{CAPM}	-0.16	0.06	-0.02	0.15	0.11	0.27
	[-3.09]	[1.14]	[-0.38]	[3.20]	[2.16]	[3.96]
α_{FF3}	-0.20	0.02	0.05	0.14	0.09	0.29
	[-3.83]	[0.45]	[0.85]	[2.92]	[1.83]	[4.22]
α_{FF4}	-0.17	0.05	0.09	0.09	0.08	0.25
	[-3.21]	[0.98]	[1.57]	[1.97]	[1.57]	[3.58]
α_{FF5}	-0.14	-0.05	0.08	0.04	0.06	0.20
	[-2.69]	[-1.07]	[1.46]	[0.84]	[1.18]	[2.96]
α_{FF6}	-0.12	-0.03	0.11	0.01	0.06	0.18
	[-2.33]	[-0.53]	[1.95]	[0.22]	[1.09]	[2.62]
Panel B: Fama and French (2018) 6-factor model loadings for RDD-sorted portfolios						
β_{MKT}	1.06	0.97	0.99	0.97	1.02	-0.04
	[87.24]	[89.57]	[75.37]	[93.04]	[84.54]	[-2.47]
β_{SMB}	0.02	-0.11	-0.01	0.00	0.04	0.03
	[0.85]	[-6.61]	[-0.27]	[0.27]	[2.44]	[1.21]
β_{HML}	0.16	0.10	-0.16	-0.03	-0.04	-0.20
	[6.75]	[4.65]	[-6.28]	[-1.26]	[-1.61]	[-6.41]
β_{RMW}	-0.04	0.16	0.01	0.14	-0.03	0.01
	[-1.61]	[7.42]	[0.26]	[6.67]	[-1.08]	[0.41]
β_{CMA}	-0.17	0.07	-0.13	0.18	0.18	0.35
	[-4.79]	[2.28]	[-3.34]	[5.81]	[5.05]	[7.53]
β_{UMD}	-0.03	-0.04	-0.04	0.04	0.01	0.03
	[-2.34]	[-3.69]	[-3.40]	[4.28]	[0.55]	[2.22]
Panel C: Average number of firms (n) and market capitalization (me)						
n	662	526	1031	582	637	
me ($\$10^6$)	1718	2811	2215	2901	1687	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the RDD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.23	0.27	0.29	0.25	0.20	0.18
			[3.27]	[3.96]	[4.22]	[3.58]	[2.96]	[2.62]
Quintile	NYSE	EW	0.16	0.18	0.18	0.17	0.16	0.16
			[3.39]	[3.85]	[3.78]	[3.69]	[3.37]	[3.41]
Quintile	Name	VW	0.24	0.29	0.31	0.25	0.24	0.21
			[3.48]	[4.16]	[4.45]	[3.66]	[3.50]	[3.01]
Quintile	Cap	VW	0.23	0.27	0.29	0.24	0.19	0.16
			[3.37]	[4.03]	[4.35]	[3.57]	[2.84]	[2.39]
Decile	NYSE	VW	0.31	0.38	0.38	0.31	0.25	0.22
			[3.35]	[4.20]	[4.15]	[3.43]	[2.82]	[2.42]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.17	0.21	0.23	0.20	0.16	0.13
			[2.46]	[3.06]	[3.27]	[2.93]	[2.27]	[2.01]
Quintile	NYSE	EW	-0.10					
			[-1.67]					
Quintile	Name	VW	0.19	0.22	0.24	0.21	0.19	0.16
			[2.68]	[3.26]	[3.48]	[3.08]	[2.73]	[2.41]
Quintile	Cap	VW	0.18	0.22	0.24	0.21	0.16	0.13
			[2.61]	[3.31]	[3.56]	[3.16]	[2.31]	[2.00]
Decile	NYSE	VW	0.24	0.31	0.30	0.27	0.20	0.17
			[2.58]	[3.31]	[3.26]	[2.89]	[2.18]	[1.90]

Table 3: Conditional sort on size and RDD

This table presents results for conditional double sorts on size and RDD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on RDD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high RDD and short stocks with low RDD .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results

		RDD Quintiles					RDD Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.63	0.98	0.97	1.09	0.70	0.07	0.09	0.08	0.06	0.08	0.08
		[2.25]	[3.62]	[3.63]	[3.81]	[2.54]	[0.91]	[1.10]	[0.96]	[0.80]	[1.02]	[0.92]
	(2)	0.74	0.94	0.84	0.93	0.80	0.06	0.10	0.07	0.07	0.04	0.05
		[2.79]	[3.73]	[3.35]	[3.77]	[3.19]	[0.79]	[1.28]	[0.86]	[0.90]	[0.55]	[0.62]
	(3)	0.74	0.83	0.85	0.89	0.84	0.10	0.15	0.16	0.13	0.14	0.12
		[3.00]	[3.75]	[3.53]	[3.90]	[3.61]	[1.31]	[1.91]	[2.08]	[1.62]	[1.83]	[1.51]
	(4)	0.70	0.82	0.83	0.80	0.84	0.15	0.17	0.17	0.14	0.13	0.11
		[3.13]	[3.86]	[3.70]	[3.82]	[3.93]	[1.92]	[2.25]	[2.20]	[1.82]	[1.62]	[1.39]
	(5)	0.42	0.65	0.60	0.70	0.71	0.29	0.33	0.34	0.28	0.21	0.18
		[2.12]	[3.62]	[3.00]	[3.87]	[3.78]	[3.18]	[3.57]	[3.69]	[3.08]	[2.32]	[1.99]

Panel B: Portfolio average number of firms and market capitalization

		RDD Quintiles Average n					RDD Quintiles Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	382	384	383	383	379	36	33	31	32	34
	(2)	106	106	106	106	106	60	60	59	60	59
	(3)	76	76	76	76	76	107	108	104	106	106
	(4)	64	64	65	65	64	230	238	229	235	228
	(5)	59	59	59	59	59	1490	2124	1805	2213	1544

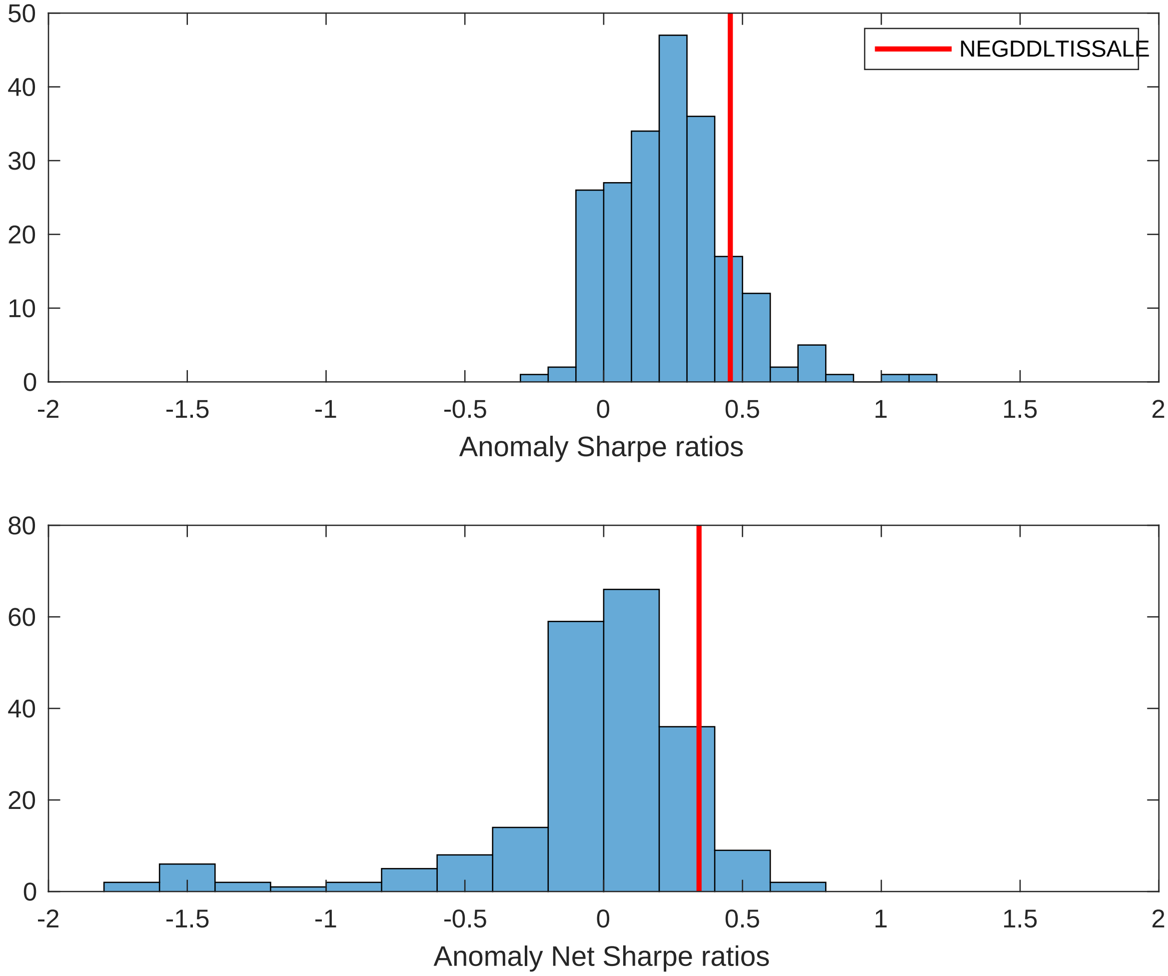


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the RDD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

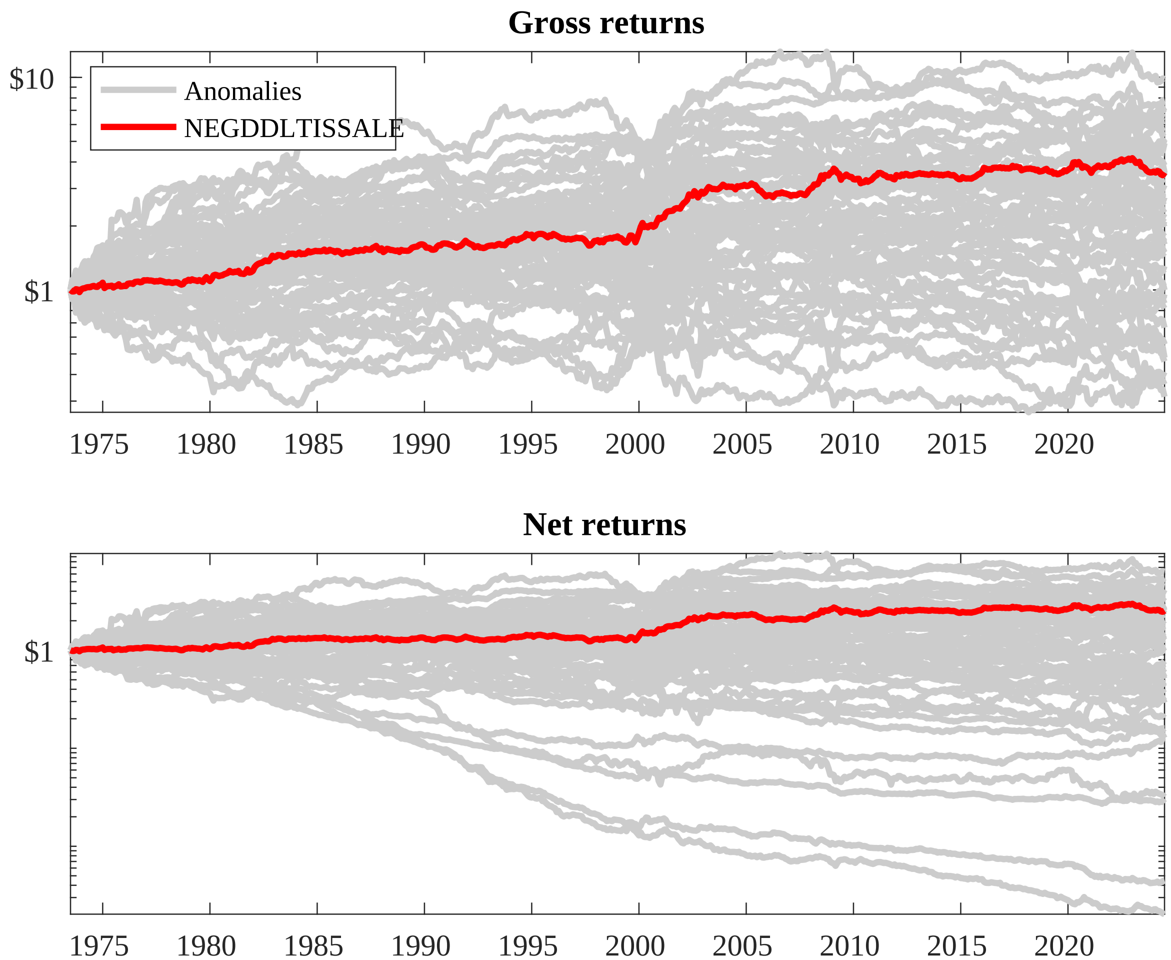
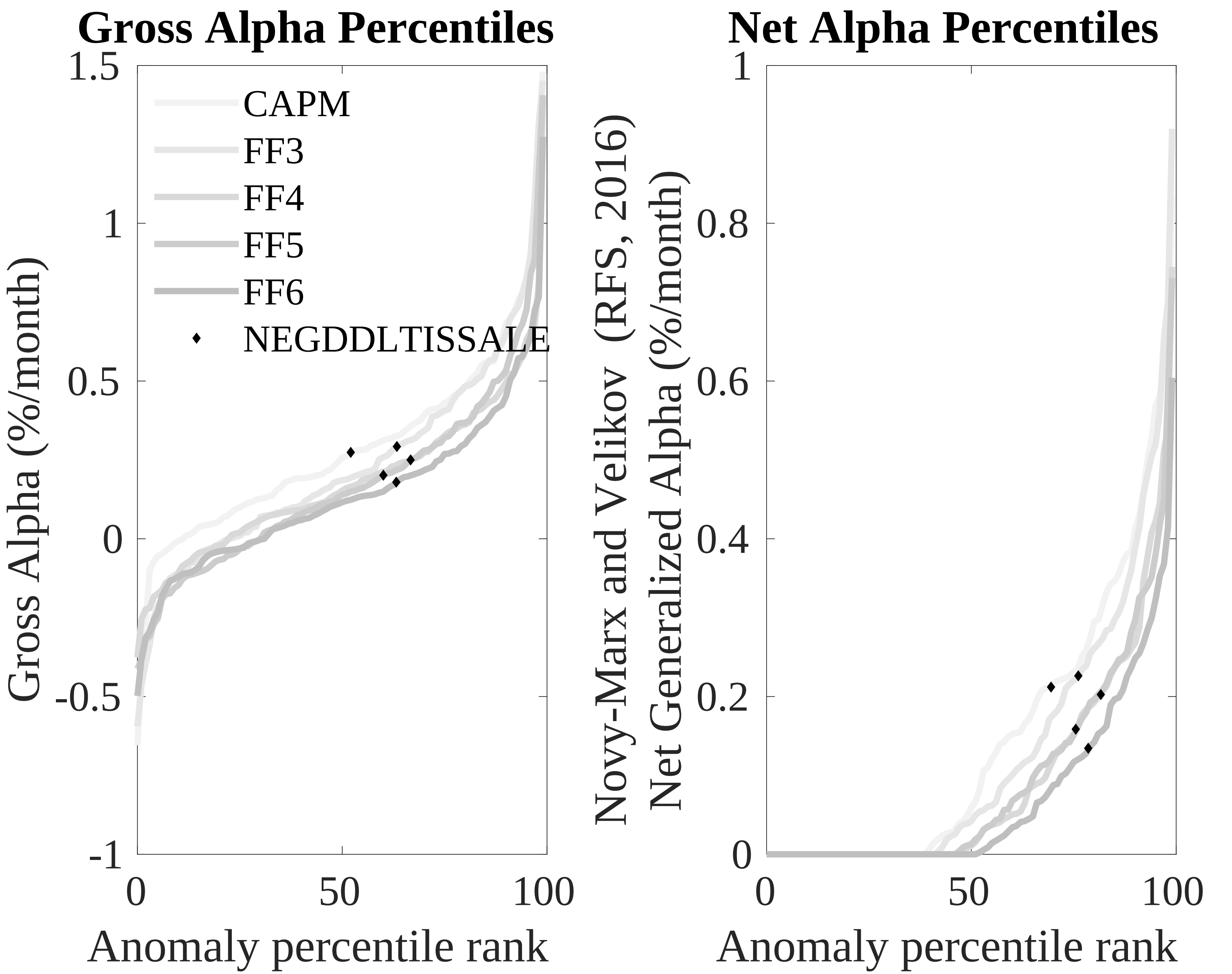
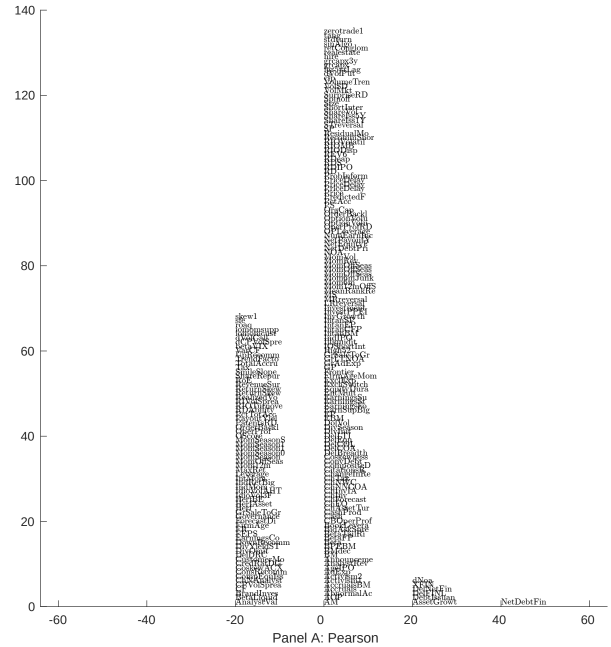


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the RDD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDDLTISSALE
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100



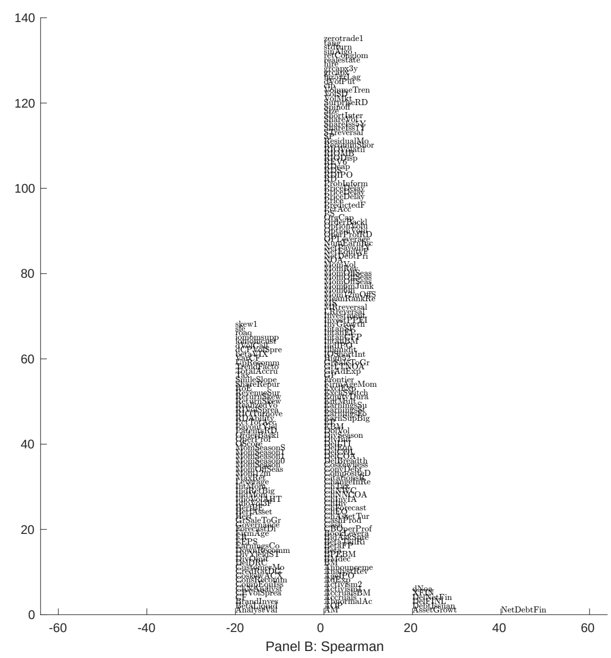


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with RDD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

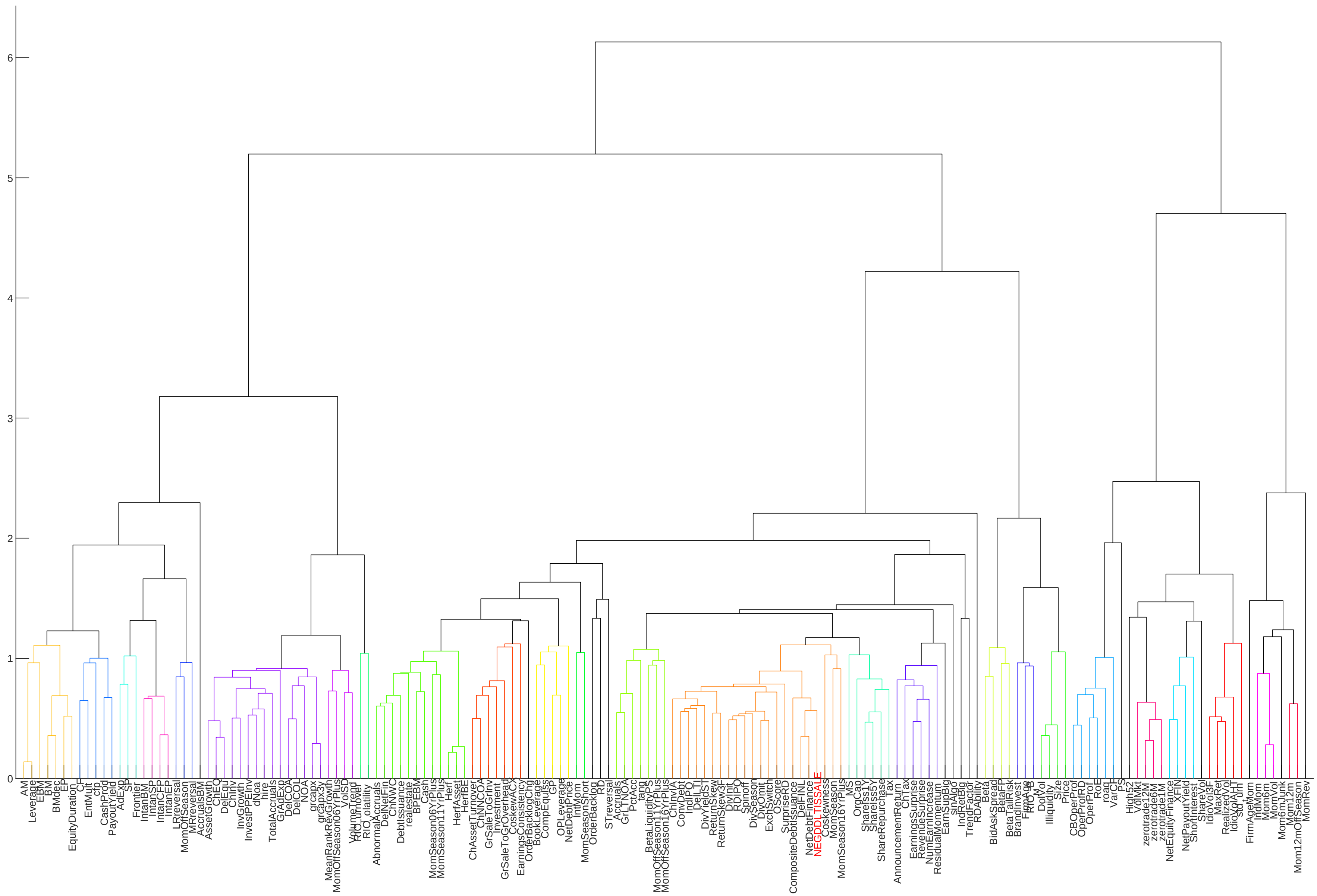


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

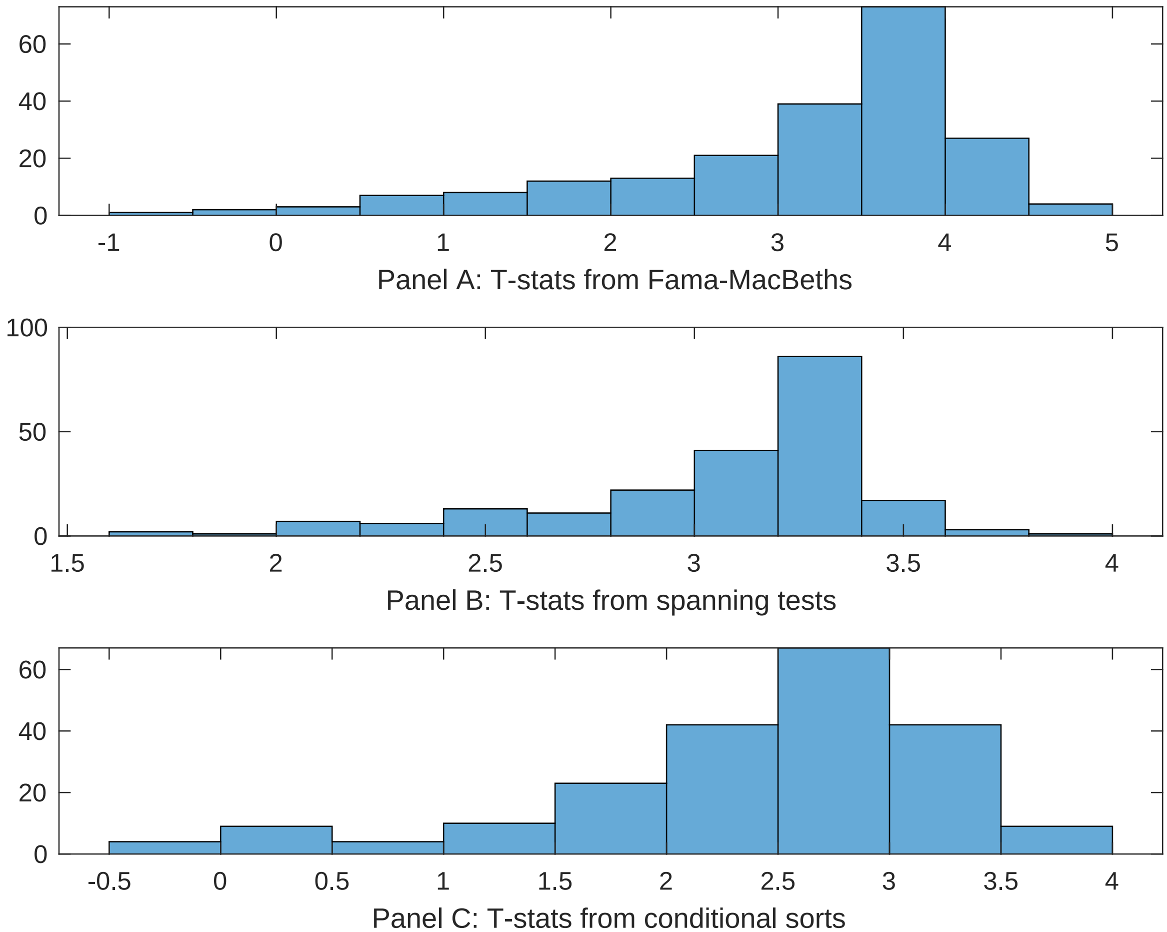


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of RDD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RDD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RDD} RDD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RDD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on RDD. Stocks are finally grouped into five RDD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RDD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on RDD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{RDD} RDD_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X, are Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13	0.13	0.14	0.14	0.14	0.13	0.15
	[5.42]	[5.46]	[5.70]	[5.32]	[5.89]	[5.42]	[5.68]
RDD	0.95	0.89	0.68	0.29	-0.13	0.19	-0.54
	[0.14]	[0.01]	[1.08]	[3.55]	[-0.19]	[2.95]	[-0.66]
Anomaly 1	0.21						0.96
	[9.63]						[1.45]
Anomaly 2		0.18					-0.53
		[9.44]					[-1.14]
Anomaly 3			0.20				0.98
			[6.75]				[1.68]
Anomaly 4				0.40			0.91
				[6.87]			[1.63]
Anomaly 5					0.11		0.76
					[8.82]		[3.66]
Anomaly 6						0.94	0.81
						[6.22]	[0.58]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the RDD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{RDD} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.15	0.15	0.15	0.17	0.17	0.19	0.15
	[2.28]	[2.22]	[2.23]	[2.54]	[2.45]	[2.73]	[2.24]
Anomaly 1	20.41						11.97
	[5.60]						[2.40]
Anomaly 2		17.17					7.28
		[4.51]					[1.45]
Anomaly 3			14.69				11.23
			[4.36]				[3.12]
Anomaly 4				6.28			4.45
				[2.37]			[1.63]
Anomaly 5					5.44		-1.81
					[1.38]		[-0.44]
Anomaly 6						6.94	3.61
						[1.89]	[0.94]
mkt	-3.56	-3.21	-1.59	-3.72	-3.61	-3.42	-1.83
	[-2.33]	[-2.07]	[-0.98]	[-2.38]	[-2.30]	[-2.18]	[-1.13]
smb	1.21	1.25	7.73	3.70	2.62	3.48	5.67
	[0.52]	[0.53]	[3.00]	[1.55]	[1.10]	[1.46]	[2.08]
hml	-18.60	-18.29	-17.80	-19.62	-19.67	-20.76	-18.02
	[-6.29]	[-6.12]	[-5.92]	[-6.50]	[-6.49]	[-6.69]	[-5.90]
rmw	-0.80	-0.10	-7.33	2.36	1.47	1.76	-6.43
	[-0.26]	[-0.03]	[-2.01]	[0.76]	[0.48]	[0.57]	[-1.76]
cma	30.13	29.67	25.85	29.83	28.82	29.23	17.62
	[6.57]	[6.31]	[5.13]	[5.80]	[4.35]	[5.19]	[2.56]
umd	2.36	2.19	3.58	3.04	3.80	3.23	1.45
	[1.53]	[1.40]	[2.33]	[1.92]	[2.43]	[2.05]	[0.92]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	18	17	16	15	14	14	20

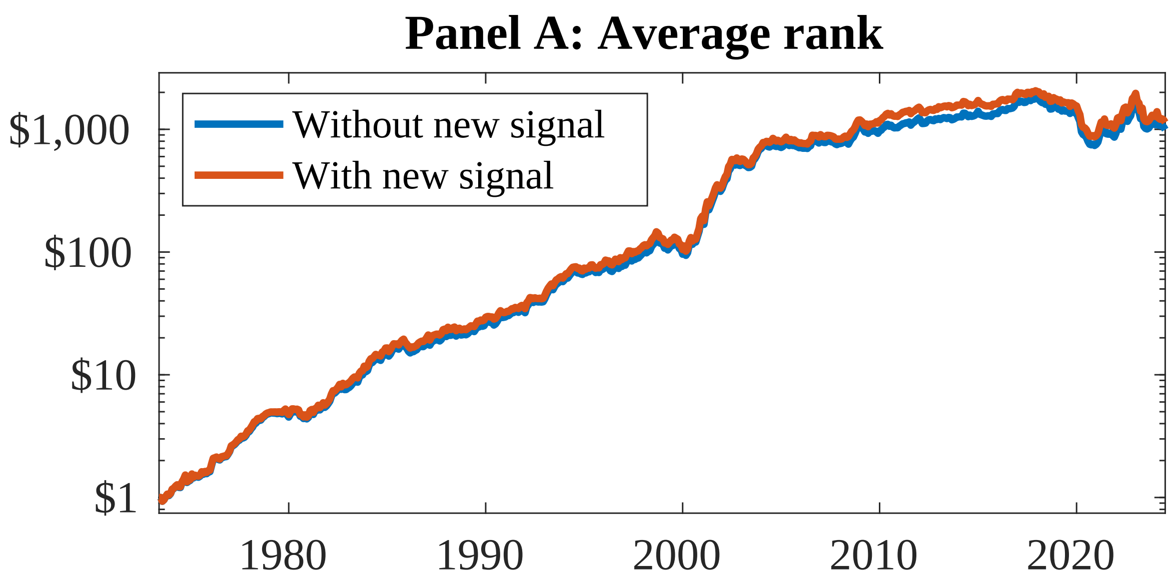


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as RDD. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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